

Snowflake Analytics Chatbot Project Documentation

Project Overview

End-to-end Data Engineering and Analytics Chatbot project built using: Azure Storage (ADLS Gen2), Databricks, Snowflake, dbt, Airflow, Power BI, Python, and Streamlit. The chatbot answers business questions from the MART_AI_DASHBOARD table without using OpenAI or Azure OpenAI APIs.

Architecture

ADLS Gen2 -> Databricks -> Snowflake -> dbt Models -> MART_AI_DASHBOARD -> Python Chatbot -> Streamlit UI.

Repository Structure

ecommerce-platform/ databricks/ snowflake/ airflow/ azure-devops/ dbt/ powerbi/ ai-chatbot/ streamlit_app.py chatbot.py snowflake_connection.py requirements.txt .env README.md

Snowflake Connection

Python connects to Snowflake using snowflake-connector-python. Data is loaded from DBT_DEV.MART_AI_DASHBOARD.

Chatbot Logic

Rule-based chatbot implementation. Supported questions: - total sales - top customer - total orders - completed orders - top product - city wise sales The chatbot returns: Question, SQL Used, and Answer.

Streamlit UI

Streamlit provides a web-based chatbot interface. Run using: streamlit run streamlit_app.py

Setup Steps

1. Create virtual environment: python -m venv env 2. Activate: env\Scripts\activate 3. Install dependencies: pip install pandas pip install snowflake-connector-python pip install python-dotenv pip install streamlit 4. Configure .env and Snowflake credentials. 5. Run Streamlit application.

Business Value

Provides self-service analytics. Demonstrates Python, Snowflake, dbt, and Streamlit integration. Suitable for portfolio, GitHub, and interview demonstrations.

Future Enhancements

- Dynamic SQL execution - Charts and dashboards - Chat history - Role-based access - Azure OpenAI integration (optional) - RAG and document search